FINAL YEAR / COLLEGE MINI PROJECT (2026)

PrepPilot

AI-Powered Resume ATS Analyzer & Interactive Mock Interview Coach

FRONTEND	BACKEND	DATABASE	AI ENGINE
React 18 + Vite	**Node.js + Express**	**MongoDB Atlas**	**Google Gemini 1.5**
Tailwind + HashRouter	RESTful MVC Architecture	Mongoose ODM + Indexes	RFC8259 Structured JSON

Presented by: **Aryan Mishra** • Deployment: `GitHub Pages + Render`

01 / THE PLACEMENT CHALLENGE

Why University Placement Candidates Struggle

75%

The ATS "Black Hole"

Over 75% of student resumes are eliminated by Automated Applicant Tracking Systems (ATS) due to missing keywords, poor formatting, or role misalignment before human review.

0×

Zero Realistic Viva Practice

Students face technical interview anxiety because traditional placement training provides theoretical lecture slides instead of interactive, multi-turn technical interview rounds.

$$$

Commercial Gatekeeping

Existing mock interview platforms (LeetCode Premium, Interviewing.io) cost $30–$150/month, making quality AI assessment inaccessible to ordinary college candidates.

🎯 **The Goal:** Provide an accessible, free, production-grade AI platform that diagnoses resumes and simulates authentic technical interviews with instant, personalized feedback.

02 / THE PREPPILOT SOLUTION

Dual-Core AI Placement Preparation Platform

01 AI Resume ATS Auditor

INSTANT PDF ANALYSIS

- ✓ **In-Memory PDF Parsing:** Extracts raw text directly from buffers with zero disk storage leaks.
- ✓ **ATS Match Score (0–100):** Evaluates keyword overlap, technical competency, and relevance against target JD.
- ✓ **Actionable Critique:** Delivers matched/missing keywords, section feedback, and 5 concrete improvement steps.

02 AI Mock Interview Coach

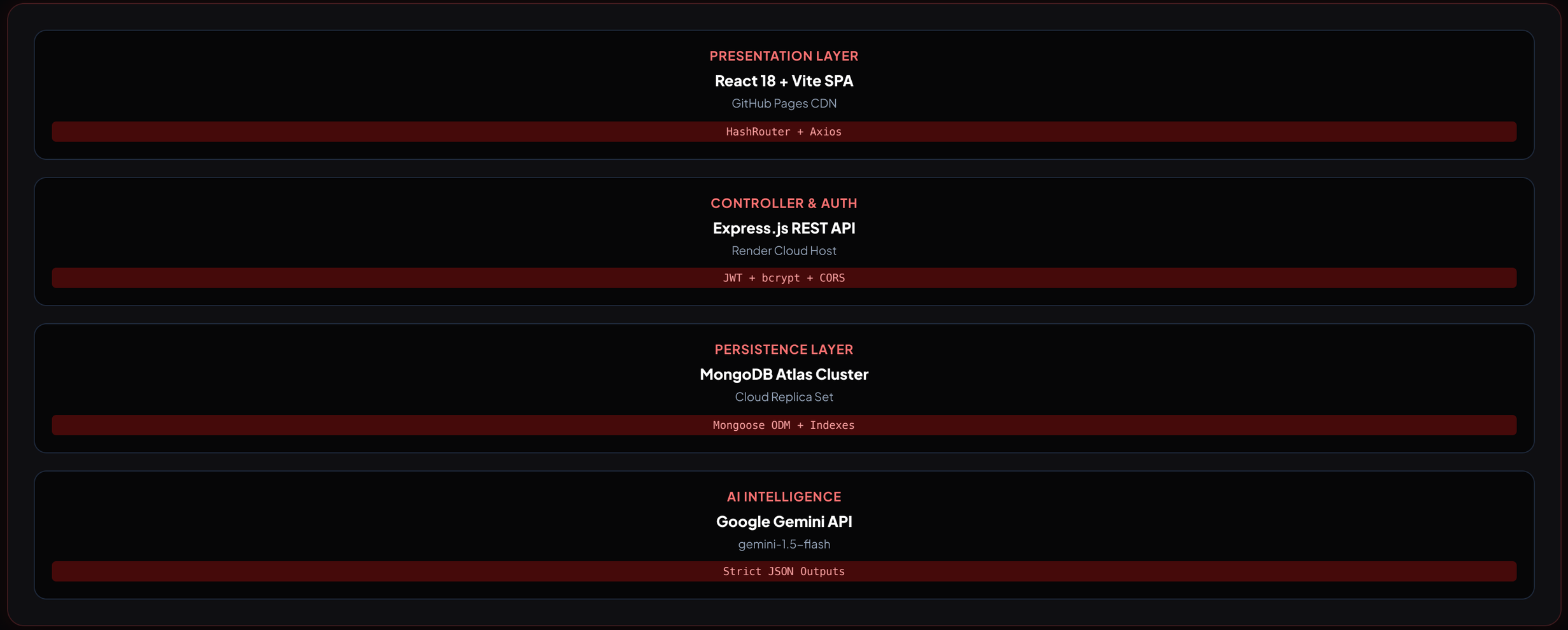
5-QUESTION TECHNICAL ROUNDS

- ✓ **Role & Difficulty Tiers:** Frontend, Backend, Full Stack, Data Science across Beginner, Intermediate, and Advanced.
- ✓ **Per-Answer AI Grading:** Evaluates answers out of 10 with constructive strengths & weaknesses critique.
- ✓ **Ideal Model Answers:** Shows industry-standard model answers for instant learning and reinforcement.

Both engines persist analyses to MongoDB Atlas and compute real-time visual progress on user dashboards.

03 / ENGINEERING DESIGN

Full-Stack MVC System Architecture



🔒 **Stateless Security:**
JWT Bearer tokens stored in localStorage, attached via Axios request interceptors.

🌐 **CORS & Proxy Isolation:**
Trust proxy enabled behind Render; origin strictly whitelisted to GitHub Pages.

⚡ **Resilient Reconnect:**
Automatic background MongoDB reconnect loop handles transient cloud cold starts.

04 / DEEP DIVE — FEATURE 1

AI Resume ATS Analyzer & PDF Exporter

ANALYSIS PIPELINE

1. **In-Memory Upload:** Multer intercepts candidate PDF in memory buffer (5MB limit, PDF-only mime filter).

2. **Text Extraction:** `pdf-parse` extracts clean text; sanitizes non-ASCII characters & line breaks.

3. **Gemini Evaluation:** Sends structured prompt comparing resume text vs. job description.

GENERATED OUTPUT SCHEMA

ATS Match Score	`0 – 100% Score`
Keyword Gap Matrix	`Matched` / `Missing`
Section Feedback	Summary, Skills, Experience, Education
Improvement Steps	5 Concrete Actionable Items
Report Export	**1-Click PDF Generation (`html2pdf`)**

💡 **Viva Highlight:** Gemini prompt enforces strict RFC8259 JSON output with custom parsing fallback logic.

`POST /api/analysis/upload`

05 / DEEP DIVE — FEATURE 2

Interactive AI Mock Interview Simulator

STEP 1: CONFIGURATION

Candidate selects target technical role and difficulty tier:

- Frontend Developer
- Backend Engineer
- Full Stack MERN
- Data Scientist
- Beginner / Intermediate / Advanced

STEP 2: DYNAMIC Q&A

AI generates 5 targeted technical questions sequentially:

- Theory + Problem Solving
- Timed answer input area
- Stateful session tracking
- Real-time question progression

STEP 3: INSTANT EVALUATION

Each submitted answer receives instant, deep analysis:

- Numerical score (/10)
- Key strengths identified
- Areas of improvement
- Ideal model answer outline

Outcome: Students experience high-pressure viva conditions and understand exactly how to structure technical answers.

`POST /api/interview/:id/answer`

06 / SECURITY & DATA INTEGRITY

Database Schemas & Production Security

🗄️ MongoDB Schemas & Relations

- **`User`**: name, email (unique index), passwordHash (bcrypt), timestamps.
- **`Analysis`**: user (ObjectId ref), atsScore, matchedKeywords, missingKeywords, suggestions.
- **`Interview`**: user (ObjectId ref), role, difficulty, questions[], averageScore.

🛡️ Security & Defensive Measures

- **Password Hashing:** `bcryptjs` pre-save hook with 10 salt rounds. Plaintext never saved.
- **JWT Verification:** Strict Bearer token authentication middleware guards private routes.
- **Rate Limiting:** `express-rate-limit` prevents API brute-forcing and Gemini cost spikes.

🔐 **Secret Isolation:** All sensitive credentials (`MONGODB_URI`, `JWT_SECRET`, `GEMINI_API_KEY`) managed via environment variables and never exposed to the client.

07 / VERIFICATION & DEVOPS

Live Cloud Deployment & Automated Testing

🧪 Automated Test Suite (Jest + Supertest)

```
PASS tests/auth.test.js (5 tests passed)
PASS tests/analysis.test.js (6 tests passed)

Test Suites: 2 passed, 2 total
Tests: 11 passed, 11 total
```

Automated test suite mocks Gemini service to validate end-to-end user flows without consuming API quotas.

☁️ Zero-Cost Production Hosting

Frontend Web App
GitHub Pages CDN (Automated Actions CI/CD)

Live Link ↗

Backend REST API
Render Node.js Cloud Web Service

Health ↗

Database Cluster
MongoDB Atlas Cloud Replica Set

Connected

🚀 **CI/CD:** GitHub Actions strictly separates `build` and `deploy` jobs to prevent artifact conflicts.

`Workflow: deploy.yml`

08 / SCALABILITY & EVOLUTION

Future Roadmap & Potential Enhancements



Voice-to-Text Mock Interviews

Integrate OpenAI Whisper / Web Speech API for real-time speech recognition, analyzing speech pace, filler words, and vocal confidence.



College TPO Analytics Portal

Admin dashboard for Training & Placement Officers to track batch-wide ATS scores, skill deficits, and candidate readiness distributions.



Live In-Browser Code Sandbox

Integrated Monaco editor with WebAssembly execution engine to evaluate coding solutions against test cases during technical rounds.

PrepPilot is architected with modular microservice-readiness, allowing new AI models or external evaluators to plug in seamlessly.

09 / PROJECT CONCLUSION

Thank You! Ready for Viva Q&A

PrepPilot successfully bridges the gap between college preparation and real-world recruitment standards through accessible, state-of-the-art AI evaluation.

LIVE APPLICATION
aryan1238.github.io/A/ ↗

BACKEND API
Render Web Service ↗

GITHUB REPO
Aryan1238/A ↗

Key Anticipated Viva Questions:

"Why Google Gemini over OpenAI?", "How is PDF parsed in memory?", "How is JWT secured?", "Why separate build & deploy jobs in GitHub Actions?"